



# Doubt Clearing Session

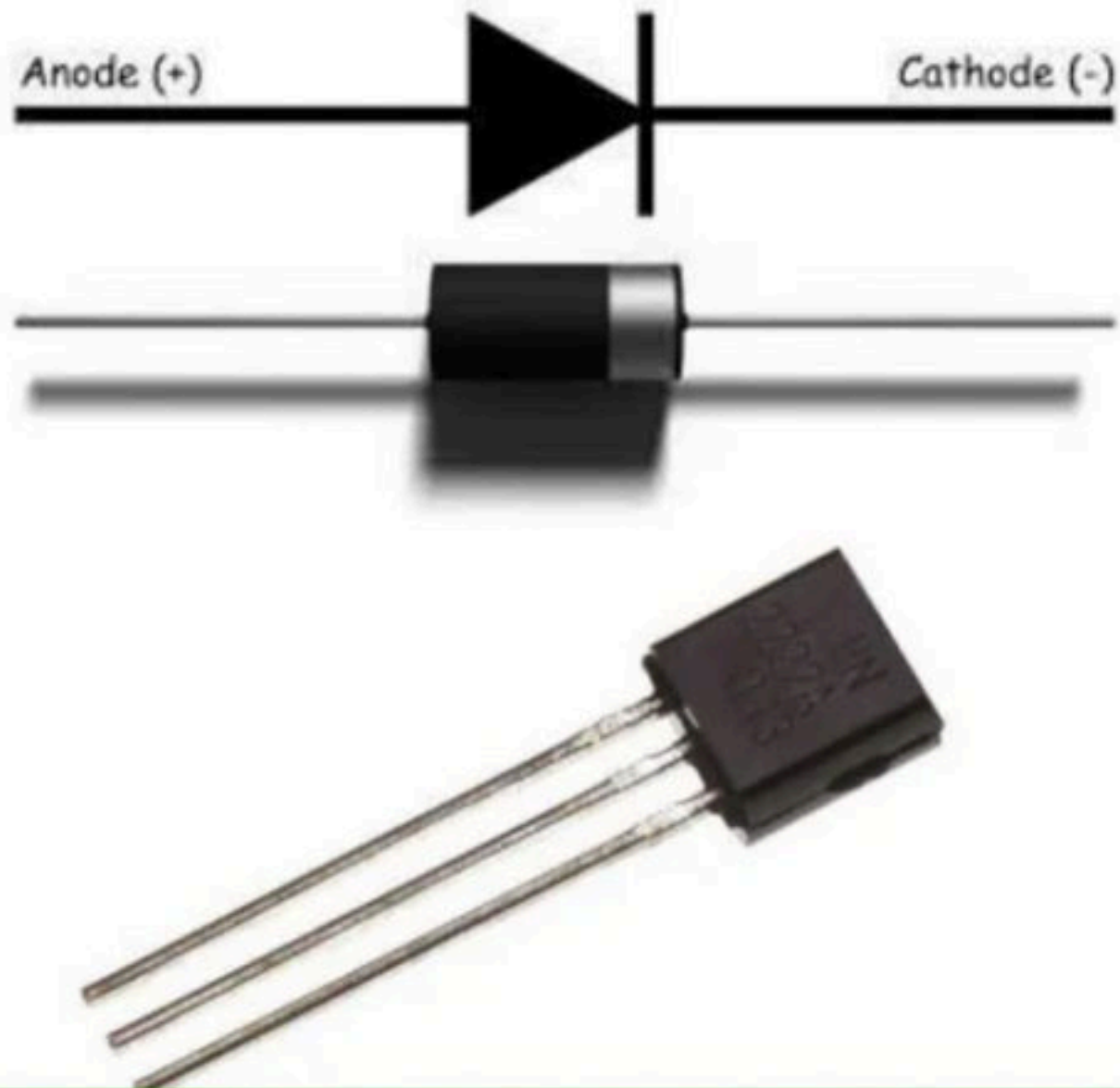
Foundation Course on Digital Electronics

## Logic gate

- In electronics, a logic gate is a physical device implementing a Boolean function
- Logic gate performs a logical operation on one or more binary inputs signals and produces a single binary output signal.
- Logic gate is the basic building block from which many kinds of logic circuits can be constructed.

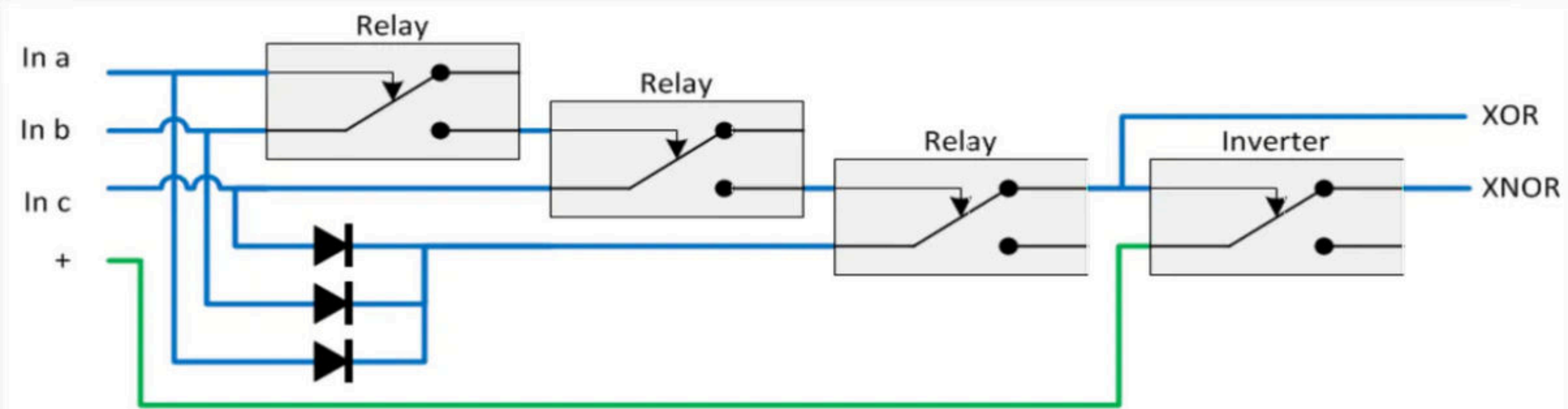
## Logic gate

- Logic gates are primarily implemented using diodes or transistors acting as electronic switches, but can also be constructed using vacuum tubes, electromagnetic relays (relay logic), fluidic logic, pneumatic logic, optics, molecules, or even mechanical elements.



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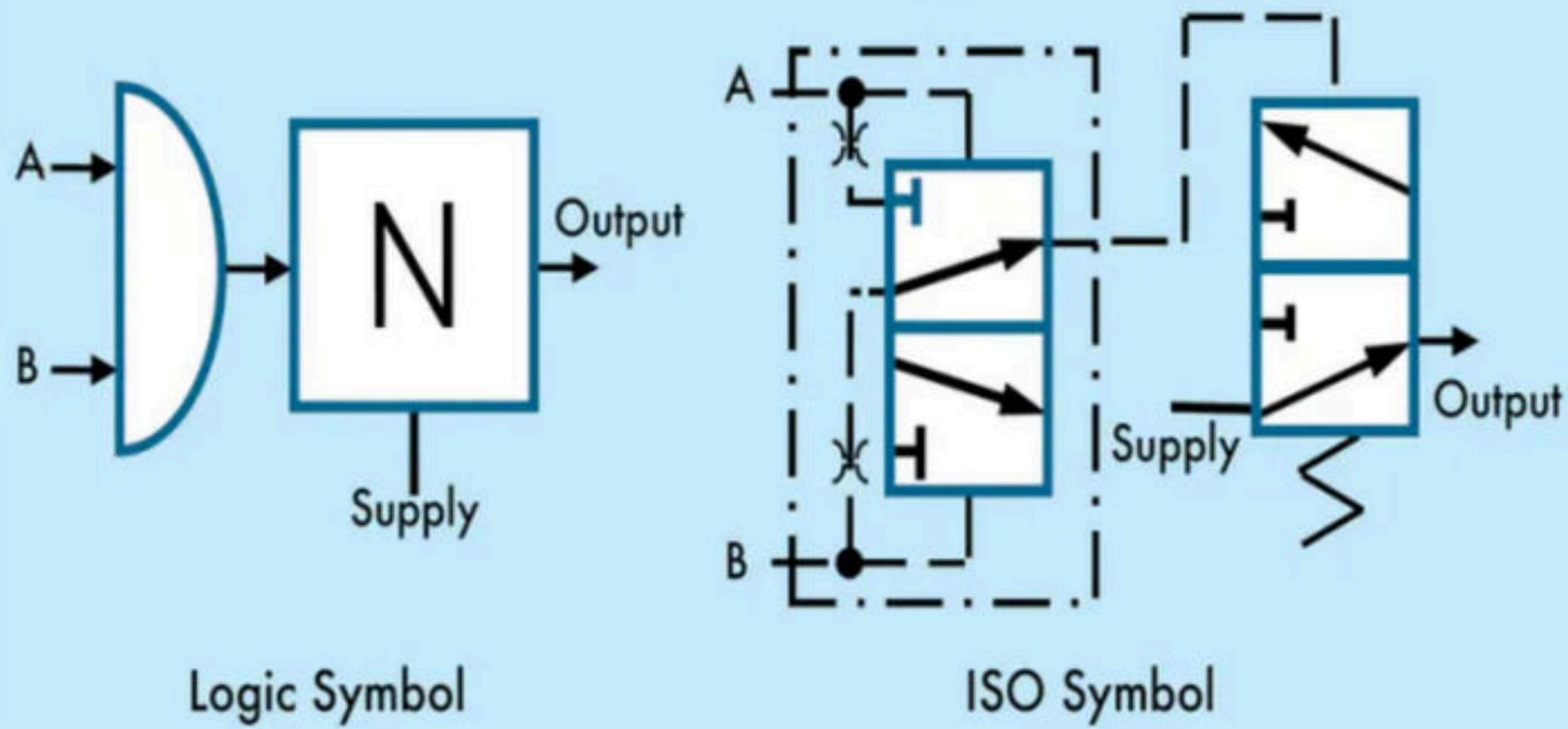




electromagnetic relays



## NAND Output

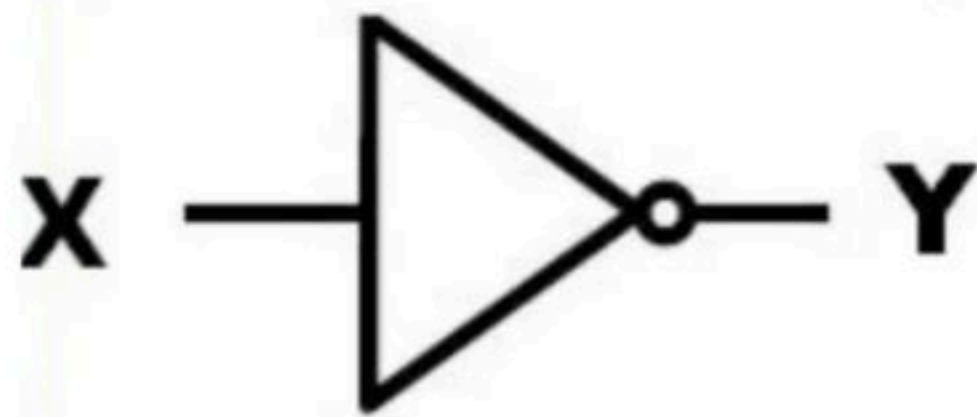


pneumatic logic

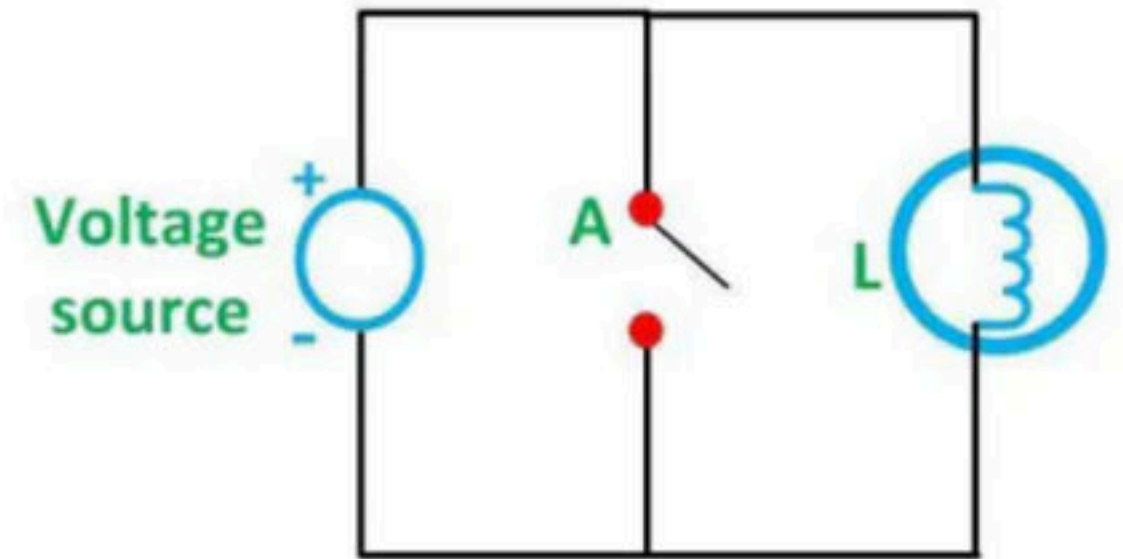
# Break

## Not Gate(Inverter)

- It represents not logical operator is also known as inverter, it is a unary operator, which simply complement the input.



| Truth Table |          |
|-------------|----------|
| Input       | Output   |
| X           | $Y = X'$ |
| 0           |          |
| 1           |          |



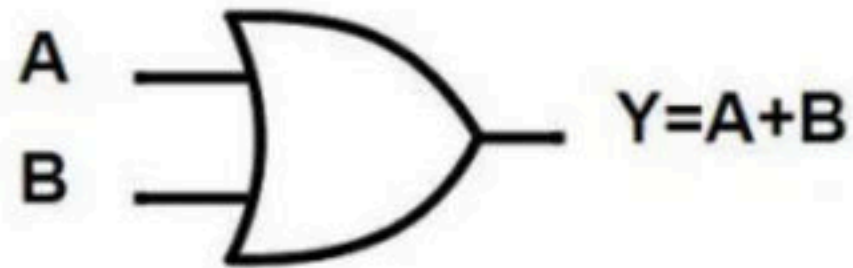
Circuit Globe



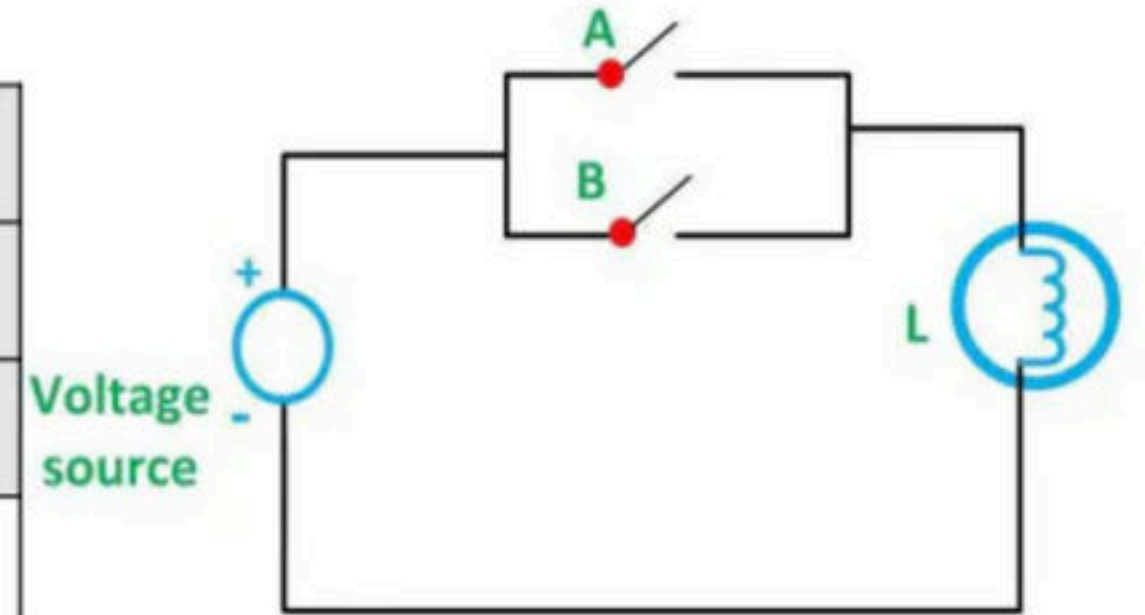
# Break

## OR Gate

- It is a digital logic gate, that implements logical disjunction. The output will be high if at least one of the input lines is high.



| Truth Table |   |             |
|-------------|---|-------------|
| Input       |   | Output      |
| A           | B | $Y = A + B$ |
| 0           | 0 |             |
| 0           | 1 |             |
| 1           | 0 |             |
| 1           | 1 |             |



Circuit Globe

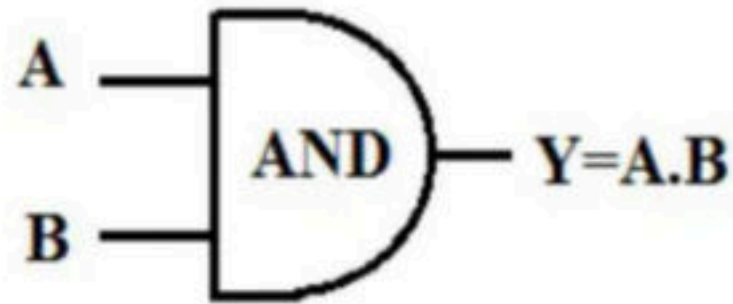
- OR gate satisfy all the 3 rules idempotent, associative, and commutative.
  1. **Idempotent Law:**  $a + a = a$
  2. **Associative law:**  $a + (b + c) = (a + b) + c$
  3. **Commutative law:**  $a + b = b + a$

# Break

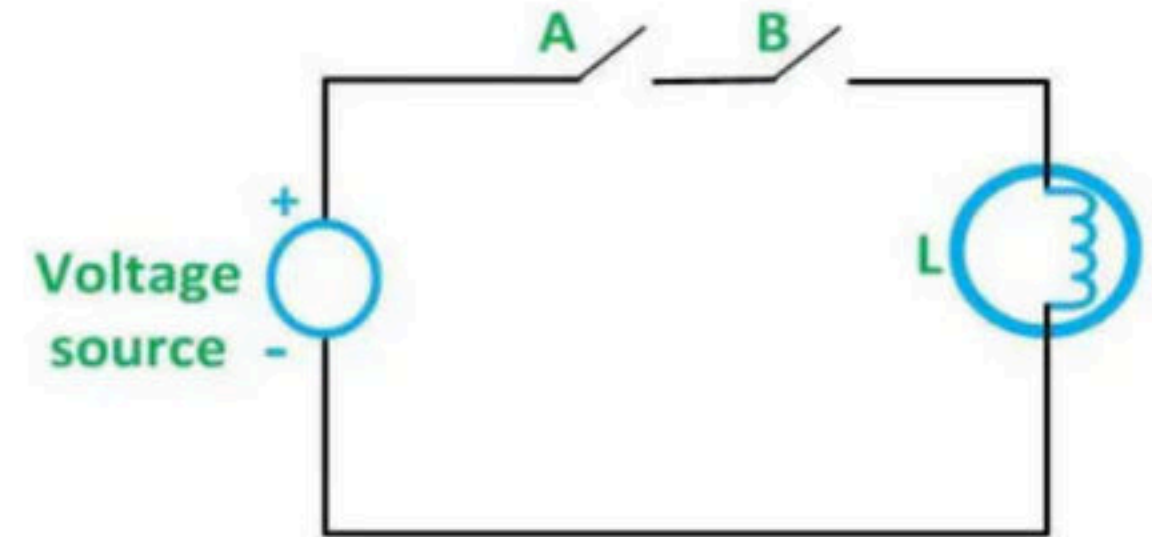


## And Gate

- It is a digital logic gate, that implements logical conjunction. Output will be high if and only if all input are high otherwise low.



| Truth Table |   |             |
|-------------|---|-------------|
| Input       |   | Output      |
| A           | B | $Y = A . B$ |
| 0           | 0 |             |
| 0           | 1 |             |
| 1           | 0 |             |
| 1           | 1 |             |



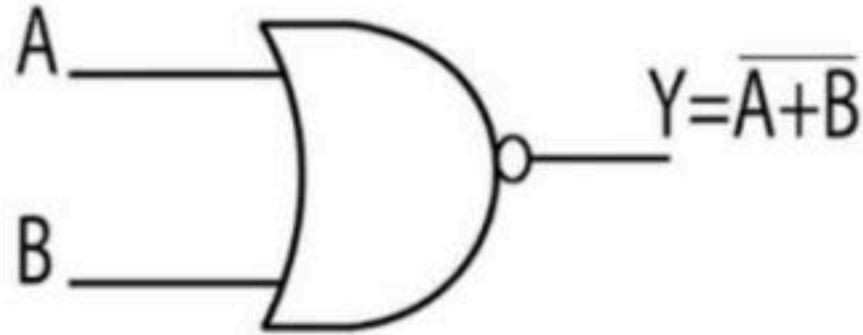
Circuit Globe

- And gate satisfy all the 3 rules idempotent, associative, and commutative.
  1. **Idempotent Law:**  $a \cdot a = a$
  2. **Associative law:**  $a \cdot (b \cdot c) = (a \cdot b) \cdot c$
  3. **Commutative law:**  $a \cdot b = b \cdot a$

# Break

## Nor gate

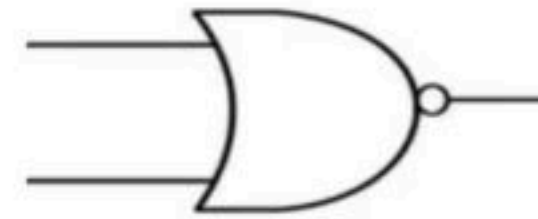
- The output will be high if and only if all inputs are low. Or simply a OR gate followed by an inverter.
- NOR gate is also called universal gate because it can be used to implement any other logic gate. We will cover this property extensively in the next chapter.



| Truth Table |   |                |
|-------------|---|----------------|
| Input       |   | Output         |
| A           | B | $Y = (A + B)'$ |
| 0           | 0 |                |
| 0           | 1 |                |
| 1           | 0 |                |
| 1           | 1 |                |

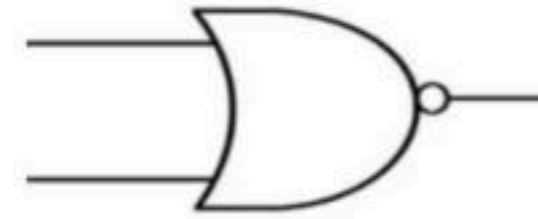
1. NOR with same gives \_\_\_\_\_

- $(a + a)' =$



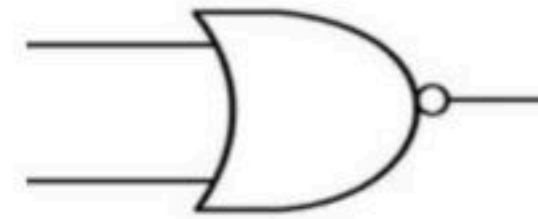
2. NOR with zero gives \_\_\_\_\_

- $(a + 0)' =$



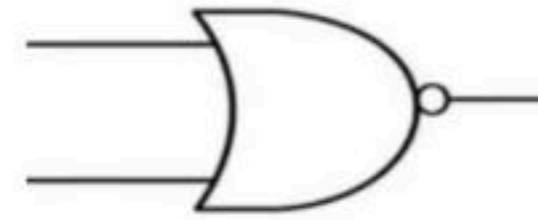
3. NOR with complement gives \_\_\_\_\_

- $(a + a')' =$



4. NOR with one gives \_\_\_\_\_

- $(a + 1)' =$





- Idempotent and associative law

- $(a + a)' \boxed{\phantom{0000}} a$

- $((a + b)' + c)' \boxed{\phantom{0000}} (a + (b + c)')'$

- Commutative law.

- $(a + b)' \boxed{\phantom{0000}} (b + a)'$

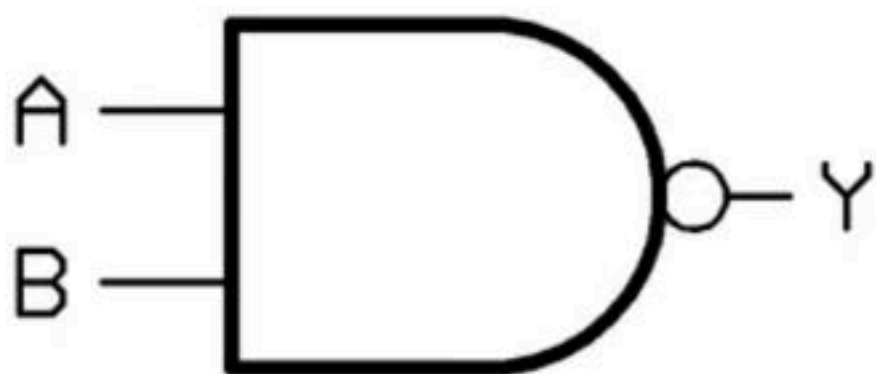
## Conclusion

- NOR with same gives complement  $\rightarrow (a + a)' = a'$
- NOR with zero gives complement  $\rightarrow (a + 0)' = a'$
- NOR with complement gives zero  $\rightarrow (a + a')' = 0$
- NOR with one gives zero  $\rightarrow (a + 1)' = 0$
  
- NOR does not satisfy idempotent and associative law
  - $(a + a)' \neq a$
  - $((a + b)' + c)' \neq (a + (b + c)')'$
- It satisfies commutative law.
  - $(a + b)' = (b + a)'$

# Break

## NAND Gate

- The output will be low if and only if all inputs are high. Or simply an and gate followed by an inverter
- NAND gate is also called universal gate because it can be used to implement any other logic gate. We will cover this property extensively in the next chapter.

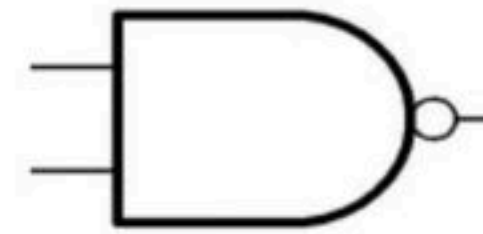


| Truth Table |   |                    |
|-------------|---|--------------------|
| Input       |   | Output             |
| A           | B | $Y = (A \cdot B)'$ |
| 0           | 0 |                    |
| 0           | 1 |                    |
| 1           | 0 |                    |
| 1           | 1 |                    |



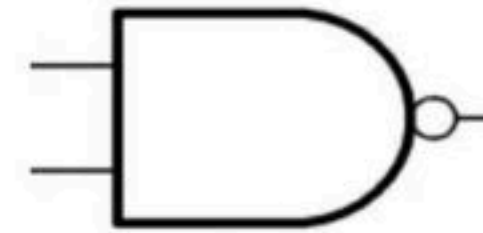
1. NAND with ZERO give\_\_\_\_\_

- $(a. 0)' =$



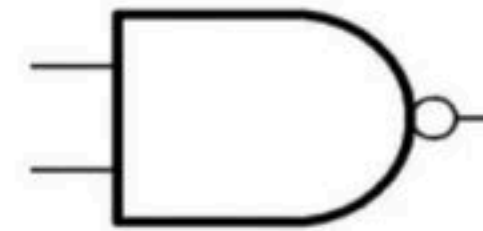
2. NAND with COMPLEMENT give\_\_\_\_\_

- $(a. a')' =$



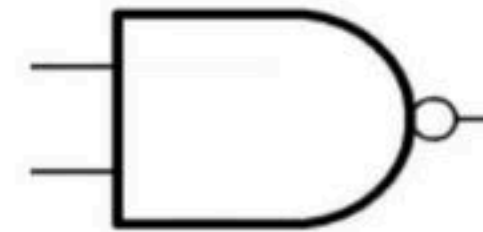
3. NAND with SAME give\_\_\_\_\_

- $(a. a)' =$



4. NAND with ONE give\_\_\_\_\_

- $(a. 1)' =$





- NAND with idempotent and associative law
  - $(a \cdot a)' \boxed{\phantom{0000}} a$
  - $((a \cdot b)' \cdot c)' \boxed{\phantom{0000}} (a \cdot (b \cdot c)')'$
- NAND with commutative law
  - $(a \cdot b)' \boxed{\phantom{0000}} (b \cdot a)'$

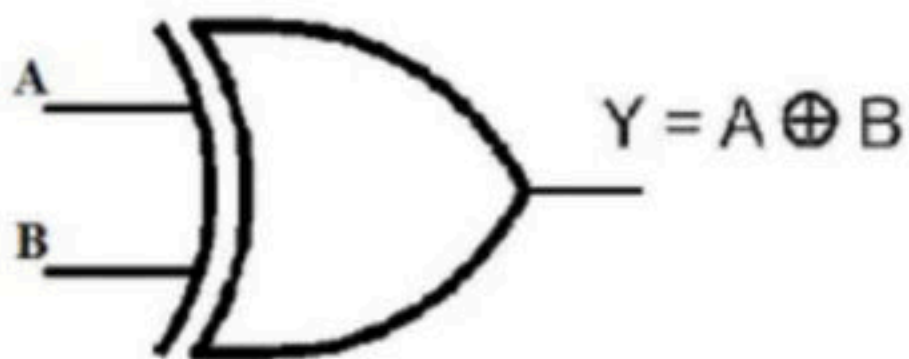
## Conclusion

- NAND with ZERO give ONE  $\rightarrow (a. 0)' = 1$
- NAND with COMPLEMENT give ONE  $\rightarrow (a. a')' = 1$
- NAND with SAME give COMPLEMENT  $\rightarrow (a. 1)' = a'$
- NAND with ONE give COMPLEMENT  $\rightarrow (a. a)' = a'$
- NAND does not satisfy idempotent and associative law
  - $(a. a)' \neq a$
  - $((a. b)'. c)' \neq (a. (b. c)')'$
- It satisfies commutative law
  - $(a. b)' = (b. a)'$

# Break

## EX-OR

- For two inputs, output will be high if and only if both the input values are different.
- $a \oplus b = a'.b + a.b'$

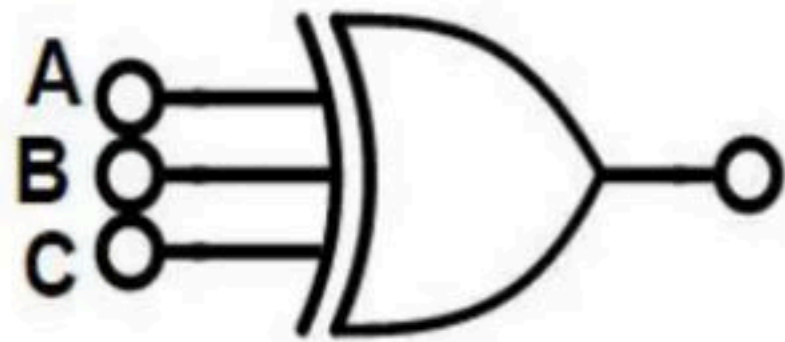


| Truth Table |   |                  |
|-------------|---|------------------|
| Input       |   | Output           |
| A           | B | $Y = A \oplus B$ |
| 0           | 0 |                  |
| 0           | 1 |                  |
| 1           | 0 |                  |
| 1           | 1 |                  |



## EX-OR

- The XOR gate is a digital logic gate that gives High as output when the number of inputs High are odd.

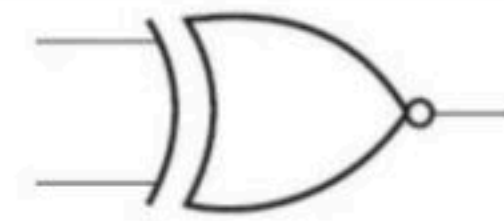


| A | B | C | $a \oplus b \oplus c$ |
|---|---|---|-----------------------|
| 0 | 0 | 0 |                       |
| 0 | 0 | 1 |                       |
| 0 | 1 | 0 |                       |
| 0 | 1 | 1 |                       |
| 1 | 0 | 0 |                       |
| 1 | 0 | 1 |                       |
| 1 | 1 | 0 |                       |
| 1 | 1 | 1 |                       |



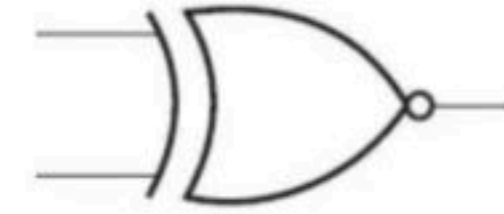
1. EX-OR with ZERO give \_\_\_\_\_

- $(a \oplus 0) =$



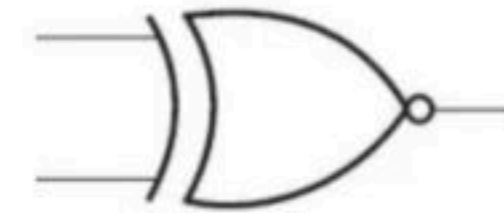
2. EX-OR with ONE give \_\_\_\_\_

- $(a \oplus 1) =$



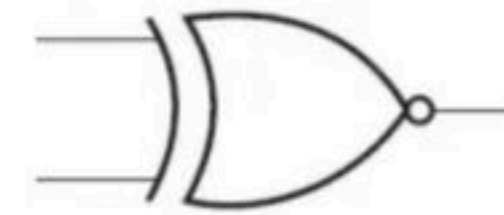
3. EX-OR with SAME give \_\_\_\_\_

- $(a \oplus a) =$



4. EX-OR with COMPLEMENT give \_\_\_\_\_

- $(a \oplus a') =$



- EX-OR with idempotent law
  - $(a \oplus a) \boxed{\phantom{000000}} a$
- EX-OR with associative and commutative law.
  - $((a \oplus b) \oplus c) \boxed{\phantom{000000}} (a \oplus (b \oplus c))$
  - $(a \oplus b) \boxed{\phantom{000000}} (b \oplus a)$

## Conclusion

- EX-OR with ZERO give SAME  $\rightarrow (a \oplus 0) = a$
- EX-OR with ONE give COMPLEMENT  $\rightarrow (a \oplus 1) = a'$
- EX-OR with SAME give ZERO  $\rightarrow (a \oplus a) = 0$
- EX-OR with COMPLEMENT give ONE  $\rightarrow (a \oplus a') = 1$
- EX-OR does not satisfy idempotent
  - $(a \oplus a) \neq a$
- It satisfies associative and commutative law.
  - $((a \oplus b) \oplus c) = (a \oplus (b \oplus c))$
  - $(a \oplus b) = (b \oplus a)$

# Break



**Q** Which of the following logic expressions is incorrect? **(NET-JULY-2016)**

a)  $1 \oplus 0 = 1$

b)  $1 \oplus 1 \oplus 1 = 1$

c)  $1 \oplus 1 \oplus 0 = 1$

d)  $1 \oplus 1 = 0$



**Q** Consider the Boolean operator with the following properties: **(GATE-2016) (1 Marks)**

$$x \# 0 = x,$$

$$x \# 1 = x',$$

$$x \# x = 0$$

$$\text{and } x \# x' = 1$$

Then  $x \# y$  is equivalent to

**a)**  $xy' + x'y$

**b)**  $xy' + x'y'$

**c)**  $x'y + xy$

**d)**  $xy + x'y'$

**Q** The binary operator # is defined by the following truth table. **(GATE-2015) (1 Marks)**

Which of the following is true about the binary operator # ?

- a) Both commutative and associative
- b) Commutative but not associative
- c) Not commutative but associative
- d) Neither commutative nor associative

| p | q | P # q |
|---|---|-------|
| 0 | 0 | 0     |
| 0 | 1 | 1     |
| 1 | 0 | 1     |
| 1 | 1 | 0     |



**Q** Let  $\oplus$  denote the Exclusive OR (XOR) operation. Let '1' and '0' denote the binary constants. Consider the following Boolean expression for F over two variables P and Q.

$$F(P, Q) = ( ( 1 \oplus P ) \oplus ( P \oplus Q ) ) \oplus ( ( P \oplus Q ) \oplus ( Q \oplus 0 ) )$$

The equivalent expression for F is

**(GATE-2014) (2 Marks)**

- (A)**  $P + Q$
- (B)**  $(P + Q)'$
- (C)**  $P \oplus Q$
- (D)**  $(P \oplus Q)'$

**Q** If  $A \oplus B = C$ , then which of the following are invalid: **(NET-JUNE-2007)**

**(A)**  $A \oplus C = B$

**(B)**  $B \oplus C = A$

**(C)**  $A \oplus B \oplus C = 1$

**(D)**  $A \oplus B \oplus C = 0$

| <b>A</b> | <b>B</b> | <b><math>A \oplus B = C</math></b> |
|----------|----------|------------------------------------|
|          |          |                                    |
|          |          |                                    |
|          |          |                                    |
|          |          |                                    |

**Q** Consider the following sequence of instructions: **(NET-DEC-2005)**

$a = a \oplus b$ ,  $b = a \oplus b$ ,  $a = b \oplus a$ . This sequence

- (A)** retains the value of the a and b
- (B)** complements the value of a and b
- (C)** swap a and b
- (D)** negates values of a and b



**Q** An example of a connective which is not associative is: **(NET-DEC-2004)**

**(A)** AND

**(B)** OR

**(C)** EX-OR

**(D)** NAND

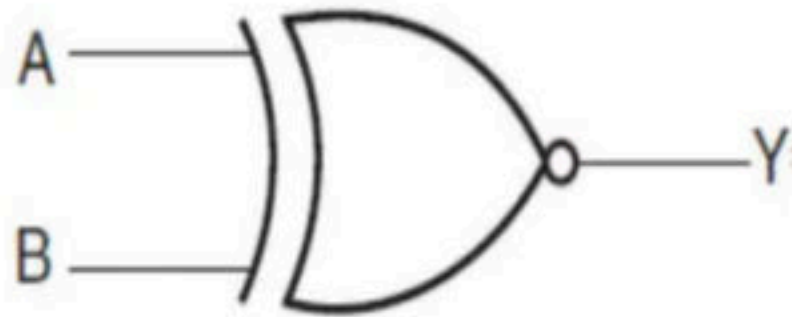
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Q  $A \oplus P \oplus A \oplus P \oplus B \oplus P \oplus B \oplus P \oplus B?$

# Break

## EX-NOR

- For two input, output will be high if and only if both the input values are same
- $a \odot b = a' \cdot b' + a \cdot b$

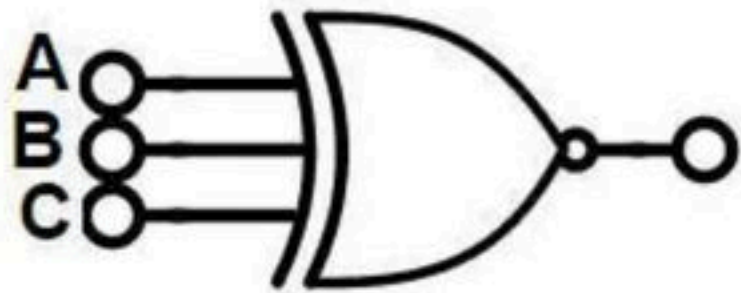


| Truth Table |   |                 |
|-------------|---|-----------------|
| Input       |   | Output          |
| A           | B | $Y = A \odot B$ |
| 0           | 0 |                 |
| 0           | 1 |                 |
| 1           | 0 |                 |
| 1           | 1 |                 |



## EX-NOR

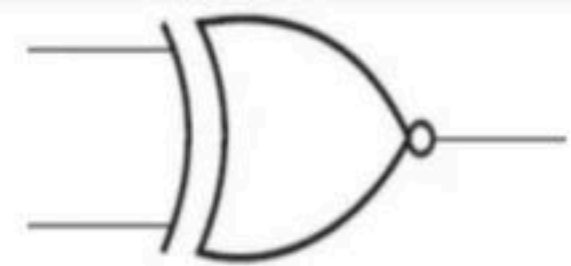
- The EX-NOR gate is a digital logic gate that gives output High when the number inputs low are even.



| A | B | C | $a \oplus b \oplus c$ |
|---|---|---|-----------------------|
| 0 | 0 | 0 |                       |
| 0 | 0 | 1 |                       |
| 0 | 1 | 0 |                       |
| 0 | 1 | 1 |                       |
| 1 | 0 | 0 |                       |
| 1 | 0 | 1 |                       |
| 1 | 1 | 0 |                       |
| 1 | 1 | 1 |                       |

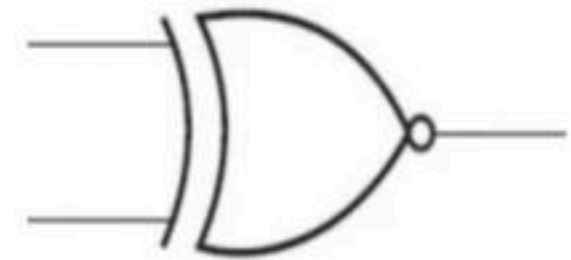
1. EX-NOR with ZERO give\_\_\_\_\_

- $(a \odot 0) =$



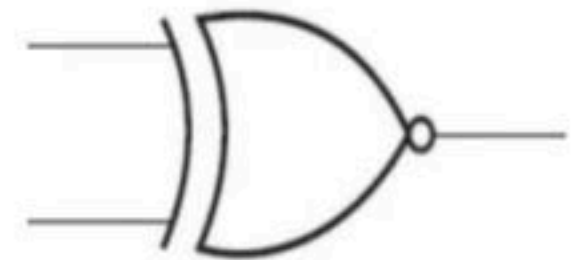
2. EX-NOR with ONE give\_\_\_\_\_

- $(a \odot 1) =$



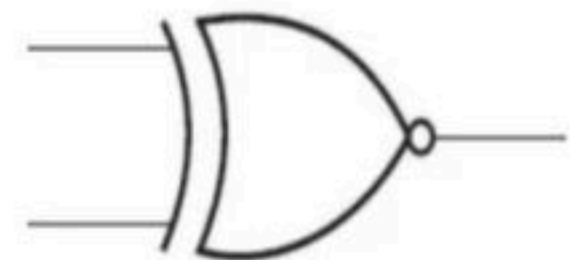
3. EX-NOR with SAME give\_\_\_\_\_

- $(a \odot a) =$



4. EX-NOR with COMPLEMENT give\_\_\_\_\_

- $(a \odot a') =$



- EX-NOR with idempotent law
  - $(a \odot a) \boxed{\phantom{000}} a$
- EX-NOR with associative and commutative law
  - $((a \odot b) \odot c) \boxed{\phantom{000}} (a \odot (b \odot c))$
  - $(a \odot b) \boxed{\phantom{000}} (b \odot a)$



## Conclusion

- EX-NOR with ZERO give COMPLEMENT  $\rightarrow (a \odot 0) = a'$
- EX-NOR with ONE give SAME  $\rightarrow (a \odot 1) = a$
- EX-NOR with SAME give ONE  $\rightarrow (a \odot a) = 1$
- EX-NOR with COMPLEMENT give ZERO  $\rightarrow (a \odot a') = 0$
  
- EX-NOR does not satisfy idempotent
  - $(a \odot a) \neq a$
- it satisfies associative and commutative law.
  - $((a \odot b) \odot c) = (a \odot (b \odot c))$
  - $(a \odot b) = (b \odot a)$



# Break

**Q** A Boolean operator  $s$  is defined as follows:

$$1 s 1 = 1,$$

$$1 s 0 = 0,$$

$$0 s 1 = 0 \text{ and}$$

$$0 s 0 = 1$$

What will be the truth value of the expression  $(x s y) s z = x s (y s z)$ ? **(NET-DEC-2013)**

- (A)** Always false
- (B)** Always true
- (C)** Sometimes true
- (D)** True when  $x, y, z$  are all true

**Q** Define the connective  $*$  for the Boolean variables  $X$  and  $Y$  as:  $X * Y = XY + X' Y'$ . Let  $Z = X * Y$ .  
**(GATE-2007) (2 Marks)**

Consider the following expressions  $P$ ,  $Q$  and  $R$ .

$P: X = Y * Z$

$Q: Y = X * Z$

$R: X * Y * Z = 1$

| $X$ | $Y$ | $X \odot Y = Z$ |
|-----|-----|-----------------|
|     |     |                 |
|     |     |                 |
|     |     |                 |
|     |     |                 |

Which of the following is TRUE?

- (A)** Only  $P$  and  $Q$  are valid
- (B)** Only  $Q$  and  $R$  are valid.
- (C)** Only  $P$  and  $R$  are valid.
- (D)** All  $P$ ,  $Q$ ,  $R$  are valid.

# Break



| A | B | C | $a \oplus b \oplus c$ | $a \odot b \odot c$ |
|---|---|---|-----------------------|---------------------|
| 0 | 0 | 0 |                       |                     |
| 0 | 0 | 1 |                       |                     |
| 0 | 1 | 0 |                       |                     |
| 0 | 1 | 1 |                       |                     |
| 1 | 0 | 0 |                       |                     |
| 1 | 0 | 1 |                       |                     |
| 1 | 1 | 0 |                       |                     |
| 1 | 1 | 1 |                       |                     |

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## Conclusion

| A | B | C | $a \oplus b \oplus c$ | $a \odot b \odot c$ |
|---|---|---|-----------------------|---------------------|
| 0 | 0 | 0 | 0                     | 0                   |
| 0 | 0 | 1 | 1                     | 1                   |
| 0 | 1 | 0 | 1                     | 1                   |
| 0 | 1 | 1 | 0                     | 0                   |
| 1 | 0 | 0 | 1                     | 1                   |
| 1 | 0 | 1 | 0                     | 0                   |
| 1 | 1 | 0 | 0                     | 0                   |
| 1 | 1 | 1 | 1                     | 1                   |

- Ex-or and ex-nor gate behaves as a complement of each other if the number of input variable is even.
- Ex-or and ex-nor gate behave same if no of input variables are odd.

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## Conclusion

1. Ex-or and ex-nor gate behaves as a complement of each other if the number of input variable is even.
2. Ex-or and ex-nor gate behave same if no of input variables are odd.

## Relations or EX-OR and EX-NOR

$$a \oplus b = a' \square b = a \square b' = (a \square b)' = (a' \square b')' = a' \square b' = (a' \square b)' = (a \square b')'$$

$$a \odot b = a' \square b = a \square b' = (a \square b)' = (a' \square b')' = a' \square b' = (a' \square b)' = (a \square b')'$$



## Conclusion

$$\begin{aligned} a \oplus b &= a' \odot b = a \odot b' = (a \odot b)' = (a' \odot b')' = a' \oplus b' = (a' \oplus b)' = (a \oplus b')' \\ a \odot b &= a' \oplus b = a \oplus b' = (a \oplus b)' = (a' \oplus b')' = a' \odot b' = (a' \odot b)' = (a \odot b')' \end{aligned}$$

**Q** Which one of the following is NOT a valid identity? **(GATE-2019) (2 Marks)**

**(A)**  $(x \oplus y) \oplus z = x \oplus (y \oplus z)$

**(B)**  $(x + y) \oplus z = x \oplus (y + z)$

**(C)**  $x \oplus y = x + y$ , if  $xy = 0$

**(D)**  $x \oplus y = (xy + x'y')'$

**Q** Let  $\oplus$  and  $\odot$  denote the Exclusive OR and Exclusive NOR operations, respectively.

Which one of the following is NOT CORRECT? **(GATE-2018) (2 Marks)**

**(A)**  $(P \oplus Q)' = P \odot Q$

**(B)**  $P \oplus Q = P \odot Q$

**(C)**  $P \oplus Q = P \oplus Q$

**(D)**  $(P \oplus P) \oplus Q = (P \odot P) \odot Q$



**Q** Let,  $X_1 \oplus X_2 \oplus X_3 \oplus X_4 = 0$  where  $X_1, X_2, X_3, X_4$  are Boolean variables, and  $\oplus$  is the XOR operator. Which one of the following must always be TRUE? **(GATE-2016)**  
**(2 Marks)**

**a)**  $X_1 X_2 X_3 X_4 = 0$

**b)**  $X_1 X_3 + X_2 = 0$

**c)**  $X'_1 \oplus X'_3 = X'_2 \oplus X'_4$

**d)**  $X_1 + X_2 + X_3 + X_4 = 0$



|    | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ | $x_1 x_2 x_3 x_4 = 0$ | $x_1 x_3 + x_2 = 0$ | $x'_1 \oplus x'_3 = x'_2 \oplus x'_4$ | $x_1 + x_2 + x_3 + x_4 = 0$ |
|----|-------|-------|-------|-------|--------------------------------------------|-----------------------|---------------------|---------------------------------------|-----------------------------|
| 0  | 0     | 0     | 0     | 0     |                                            |                       |                     |                                       |                             |
| 1  | 0     | 0     | 0     | 1     |                                            |                       |                     |                                       |                             |
| 2  | 0     | 0     | 1     | 0     |                                            |                       |                     |                                       |                             |
| 3  | 0     | 0     | 1     | 1     |                                            |                       |                     |                                       |                             |
| 4  | 0     | 1     | 0     | 0     |                                            |                       |                     |                                       |                             |
| 5  | 0     | 1     | 0     | 1     |                                            |                       |                     |                                       |                             |
| 6  | 0     | 1     | 1     | 0     |                                            |                       |                     |                                       |                             |
| 7  | 0     | 1     | 1     | 1     |                                            |                       |                     |                                       |                             |
| 8  | 1     | 0     | 0     | 0     |                                            |                       |                     |                                       |                             |
| 9  | 1     | 0     | 0     | 1     |                                            |                       |                     |                                       |                             |
| 10 | 1     | 0     | 1     | 0     |                                            |                       |                     |                                       |                             |
| 11 | 1     | 0     | 1     | 1     |                                            |                       |                     |                                       |                             |
| 12 | 1     | 1     | 0     | 0     |                                            |                       |                     |                                       |                             |
| 13 | 1     | 1     | 0     | 1     |                                            |                       |                     |                                       |                             |
| 14 | 1     | 1     | 1     | 0     |                                            |                       |                     |                                       |                             |
| 15 | 1     | 1     | 1     | 1     |                                            |                       |                     |                                       |                             |

Q Which one of the following expressions does NOT represent exclusive NOR of x and y? (GATE-2013) (1 Marks)

(A)  ~~$xy + x'y'$~~

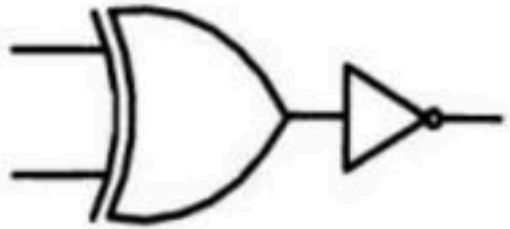
(B)  ~~$x \wedge y'$  where  $\wedge$  is XOR~~

(C)  ~~$x' \wedge y$  where  $\wedge$  is XOR~~

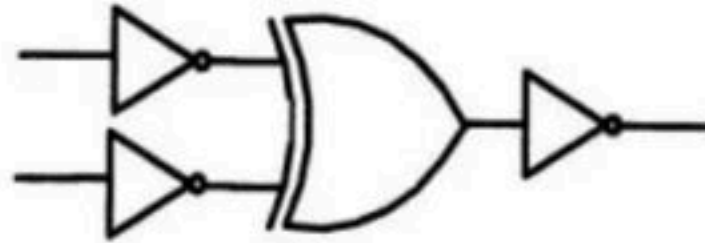
(D)  $x'$   $\wedge$   $y'$  where  $\wedge$  is XOR

**Q** Which one of the following circuits is NOT equivalent to a 2-input XNOR (exclusive NOR) gate? **(GATE-2011) (1 Marks)**

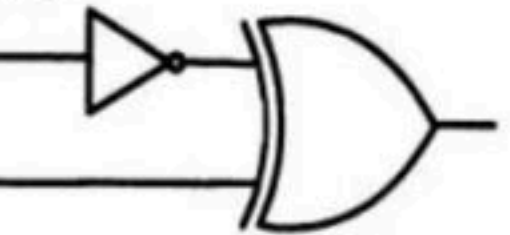
(A)



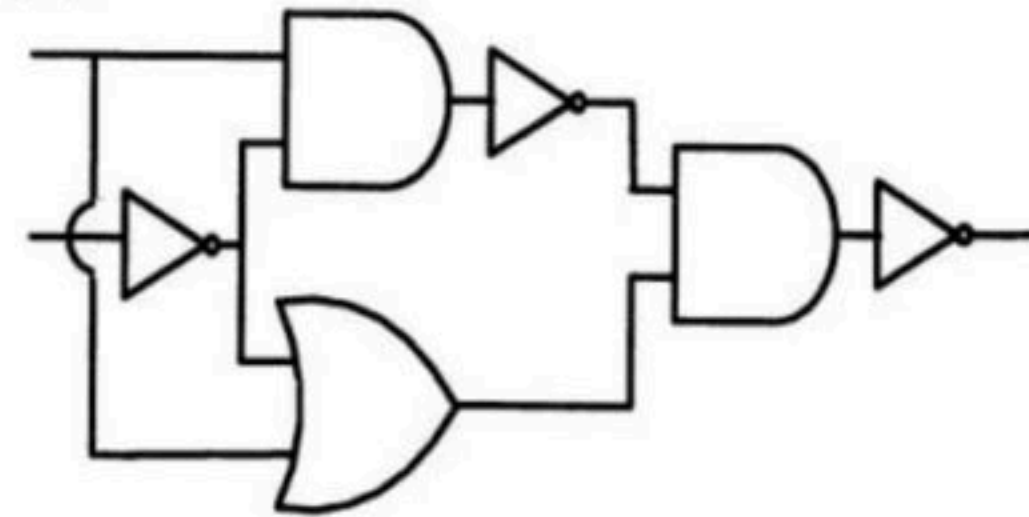
(B)



(C)



(D)





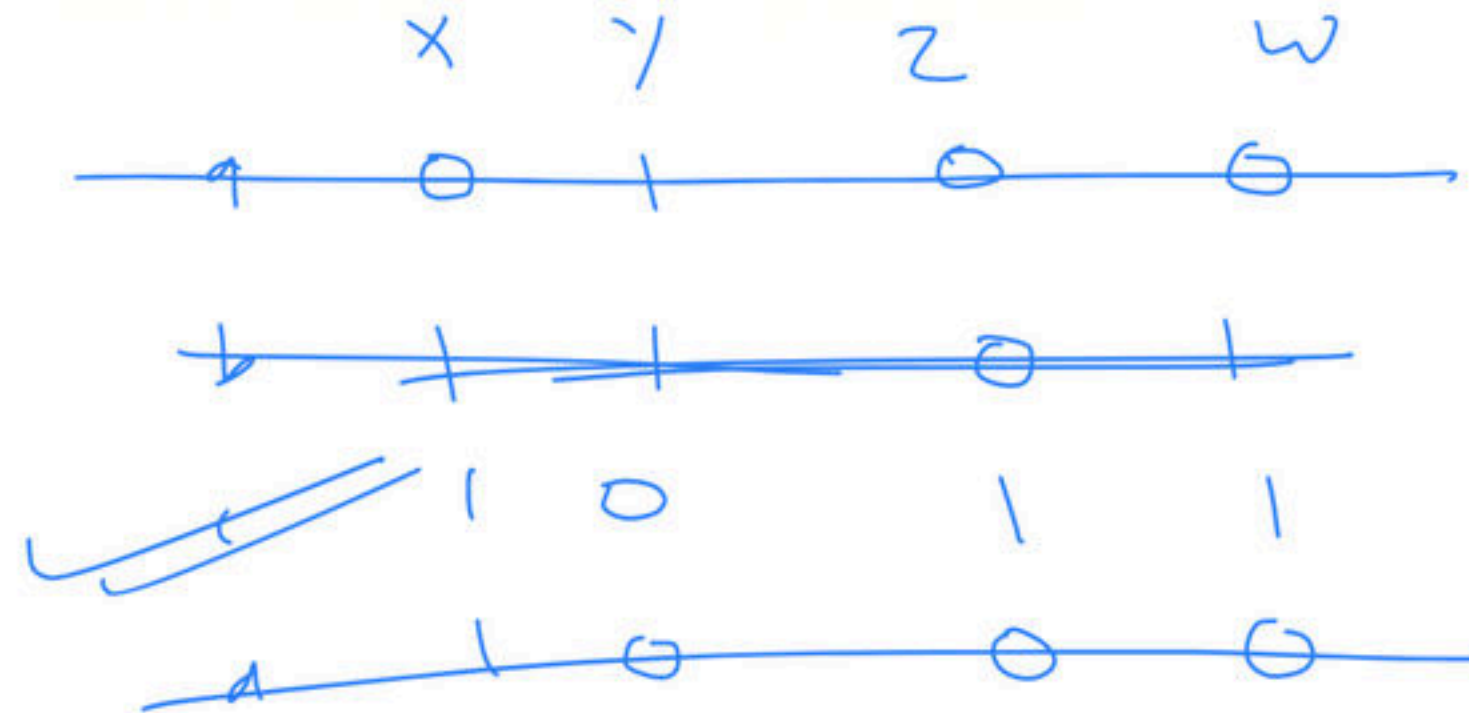
**Q** The simultaneous equations on the Boolean variables  $x, y, z$  and  $w$ ,

$$x + y + z = 1$$

$$xy = 0$$

$$xz + w = 1$$

$$xy + z'w' = 0$$



have the following solution for  $x, y, z$  and  $w$ , respectively. **(GATE-2000) (2 Marks)**

**(A)** 0 1 0 0

**(B)** 1 1 0 1

**(C)** 1 0 1 1

**(D)** 1 0 0 0



## Boolean Expressions

- Boolean expressions are the method using which we save the information about the Boolean function, that when we get value 1 and when we get value 0 as output.
- So, we convert the truth table of the function into an expression, reading which we can understand where the output is 1 and when it is zero.

| Light | Day | Engine | Warning |
|-------|-----|--------|---------|
| 0     | 0   | 0      | 0       |
| 0     | 0   | 1      | 1       |
| 0     | 1   | 0      | 0       |
| 0     | 1   | 1      | 0       |
| 1     | 0   | 0      | 1       |
| 1     | 0   | 1      | 0       |
| 1     | 1   | 0      | 1       |
| 1     | 1   | 1      | 1       |

•  $W = a'b'c + ab'c' + abc' + abc$

- There are two popular approaches for writing these expressions.
  - Sum of Product (SOP) (which remember when we get 1)
  - Product of Sum (POS) (which remember when we get 0)
- Make a note of this as we are studying Boolean function remembering both 0 and 1 is not required, so we can either concentrate on 0 or 1.

| Light | Day | Engine | Warning |
|-------|-----|--------|---------|
| 0     | 0   | 0      | 0       |
| 0     | 0   | 1      | 1       |
| 0     | 1   | 0      | 0       |
| 0     | 1   | 1      | 0       |
| 1     | 0   | 0      | 1       |
| 1     | 0   | 1      | 0       |
| 1     | 1   | 0      | 1       |
| 1     | 1   | 1      | 1       |

$$W(L, D, E) = \sum_m (1, 4, 6, 7)$$

$$W(L, D, E) = \prod_M (0, 2, 3, 5)$$



- Power of SOP and POS both are same, i.e. any Boolean functions can be represented using both SOP and POS.
- Mathematically speaking we should choose (0 or 1), whatever is less in number because then it will be easy to represent

| A | B | $f_1 = A + B$ | $f_2 = A \cdot B$ |
|---|---|---------------|-------------------|
| 0 | 0 | 0             | 0                 |
| 0 | 1 | 1             | 0                 |
| 1 | 0 | 1             | 0                 |
| 1 | 1 | 1             | 1                 |

**Q The truth table represents the Boolean function (GATE-2012) (1 Marks)**

**(A)  $X$**

**(B)  $X+Y$**

**(C)  $X \oplus Y$**

**(D)  $Y$**

| X | Y | F (X, Y) |
|---|---|----------|
| 0 | 0 | 0        |
| 0 | 1 | 0        |
| 1 | 0 | 1        |
| 1 | 1 | 1        |



# Break

## SOP (sum of product)

- A sum of product form expression contains product terms (AND terms) which are sum (OR) together, that's why called sum of product.
- Each product terms (AND terms) consists of one or more literals (variables) appearing either in complements or uncomplemented form. E.g.  $a'b + b'c' + abc$

- A product term which contains all the literals (variables) either in complemented or uncomplemented form is called minterm.
- In a n variable function, there will be  $2^n$  minterms.

$$f(a, b, c) = a \cdot \bar{b}$$

| Binary Representation | Sequence | Minterm  | Designation |
|-----------------------|----------|----------|-------------|
| 000                   | 0        | $a'b'c'$ | $m_0$       |
| 001                   | 1        | $a'b'c$  | $m_1$       |
| 010                   | 2        | $a'bc'$  | $m_2$       |
| 011                   | 3        | $a'bc$   | $m_3$       |
| 100                   | 4        | $ab'c'$  | $m_4$       |
| 101                   | 5        | $ab'c$   | $m_5$       |
| 110                   | 6        | $abc'$   | $m_6$       |
| 111                   | 7        | $abc$    | $m_7$       |



- The result of a product term must always be 1, If a literal is having value 1 then it is ok, but if not, then we complement those which is 0, it to make it 1.  

- There is only 1 input sequence of variables for any minterm on which the output is 1, so it represents information.  

- Then the sum of all product term(minterm) to form a function, and functions will have value 1, if at least one of the product term(minterm) is 1.

| Binary Representation | Sequence | Minterm  | Designation |
|-----------------------|----------|----------|-------------|
| 000                   | 0        | $a'b'c'$ | $m_0$       |
| 001                   | 1        | $a'b'c$  | $m_1$       |
| 010                   | 2        | $a'bc'$  | $m_2$       |
| 011                   | 3        | $a'bc$   | $m_3$       |
| 100                   | 4        | $ab'c'$  | $m_4$       |
| 101                   | 5        | $ab'c$   | $m_5$       |
| 110                   | 6        | $abc'$   | $m_6$       |
| 111                   | 7        | $abc$    | $m_7$       |

$$f(a, b, c) = \bar{a}\bar{b}\bar{c} + \bar{a}\bar{b}c + \bar{a}b\bar{c} + \bar{a}bc$$

$$f(a, b, c) = m_1 + m_4 + m_6 + m_7$$

$$f(a, b, c) = \sum m(1, 4, 6, 7)$$



## Canonical logic forms

- Either in POS or SOP form it is not essential that all product or sum terms contains all the literals.
- **Canonical SOP form**: - In a sum of product form expression, if each AND term (product term) consists all the literals(variables) appearing either in complements or uncomplemented form. E.g.  $a'bc + ab'c' + abc$ . Then the form is said to be canonical SOP.

Sol  $\rightarrow$

$$f(a, b, c) = \overline{a}b + \overline{b}c + ab$$

$\downarrow$

$$ab(\overline{c} + c)$$
$$\underline{ab\overline{c}} + \underline{abc}$$
$$\overline{a}\overline{b}\overline{c} + \overline{a}\overline{b}c + \underline{a\overline{b}\overline{c}} + \underline{abc}$$

**Q** The minterm expansion of  $f(P, Q, R) = PQ + QR' + PR'$  is **(GATE-2010) (2 Marks)**

$$PQ(R + \bar{R}) + (\bar{P} + P)Q\bar{R} + \overbrace{P\bar{Q}\bar{R}}^{(0+0)}$$

$$\cancel{PQR} + \cancel{PQ\bar{R}} + \cancel{\bar{P}QR} + \cancel{PQ\bar{R}} + \cancel{P\bar{Q}R} + \cancel{PQ\bar{R}}$$

~~(A)  $m_2 + m_4 + m_6 + m_7$~~

(B)  $m_0 + m_1 + m_3 + m_5$

(C)  $m_0 + m_1 + m_6 + m_7$

(D)  $m_2 + m_3 + m_4 + m_5$

$$f(P, Q, R) = PQ + Q\bar{R} + P\bar{R}$$



2967



**Q** Consider a function and find no of minterms  $F(a, b, c) = a + a'b + abc + bc'$ ?

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# Break



## POS (Product of Sum)

- A product of sum (POS) form of expression contains OR (sum) terms which are AND (product) together, that's why called product of sum expression.
- Each OR term (sum term) consists of one or more literals(variables) appearing either in complemented or uncomplemented form.  $(a' + b) \cdot (b' + c') \cdot (a + c)$

- A OR (sum) term which contains all the literals(variables) either in complemented or uncomplemented form is called maxterm. In a n variable function, there will be  $2^n$  maxterms.

$$(\overline{a} + \overline{b} + c)$$

1        1        0

$$0 \quad 1 \quad \cancel{0}$$

$$a + \overline{b} + \overline{c}$$

$$a + \overline{b} + c = 0$$

↓        ↓        ↓

0        1        0

| Binary Representation | Sequence | <u>Maxterm</u> | Designation |
|-----------------------|----------|----------------|-------------|
| 000                   | 0        | $a + b + c$    | $M_0$       |
| 001                   | 1        | $a + b + c'$   | $M_1$       |
| 010                   | 2        | $a + b' + c$   | $M_2$       |
| 011                   | 3        | $a + b' + c'$  | $M_3$       |
| 100                   | 4        | $a' + b + c$   | $M_4$       |
| 101                   | 5        | $a' + b + c'$  | $M_5$       |
| 110                   | 6        | $a' + b' + c$  | $M_6$       |
| 111                   | 7        | $a' + b' + c'$ | $M_7$       |

$$f(a,b,c) = (a + b + c) \cdot (a + \bar{b} + c) \cdot (a + \bar{b} + \bar{c}) \cdot (\bar{a} + b + \bar{c})$$

0   0   0
0   1   0
0   1   1
1   0   1

$$f(a,b,c) = M_0 \cdot M_2 \cdot M_3 \cdot M_5$$

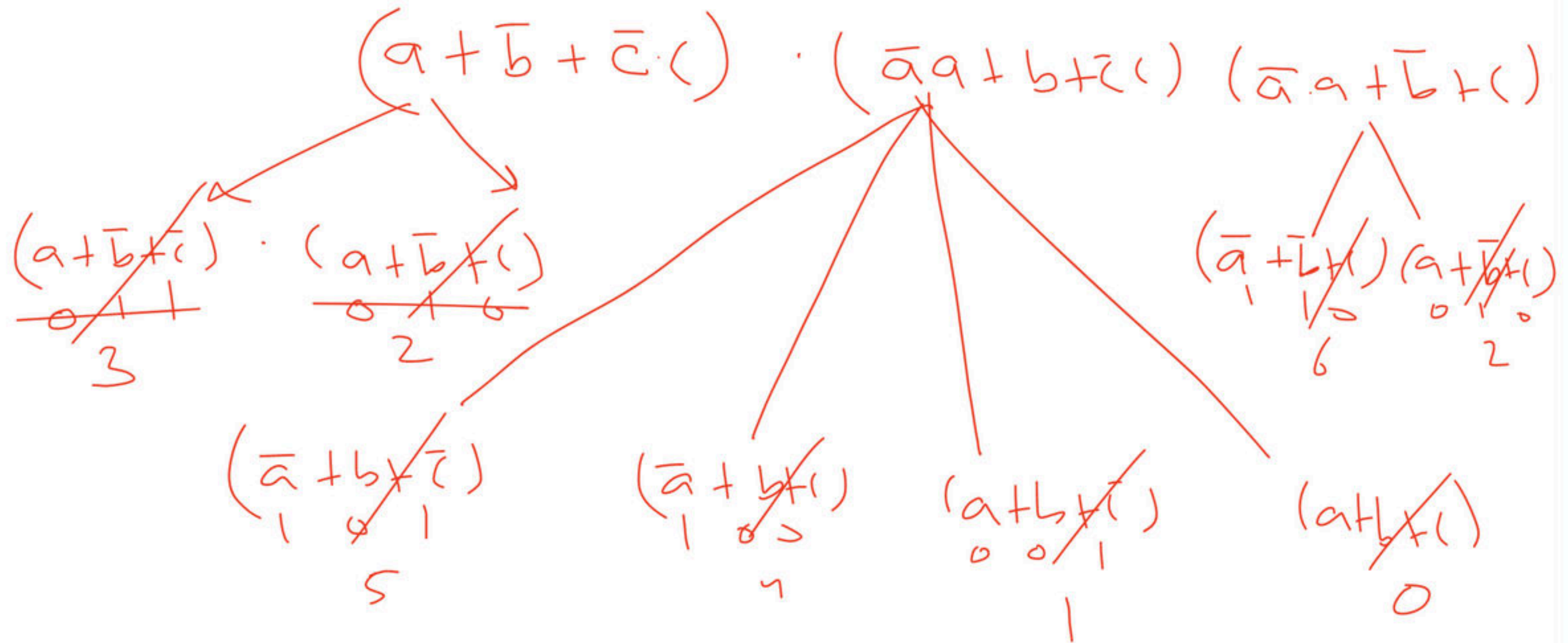
$$f(a,b,c) = \Pi_M(0, 2, 3, 5)$$

$$\Sigma ( ) \cdot ( ) \cdot ( ) \cdot ( )$$



pos  $\rightarrow f(a,b,c) = \underbrace{(a + \bar{b})} \cdot \underbrace{(b)} \cdot \underbrace{(\bar{b} + c)}$

$\Lambda_m(0,1,2,3,4,5,6) = \Sigma_m(7) \leftarrow$





~~T~~ T

~~F~~ F

Sop pos

~~a + b~~

88%

Soy

105

~~(a + b)~~ ~~(c + d)~~

12%

- The result of the OR (sum) term must be 0, so If a variable is having value 0 then it is ok, but if not then we complement the variable it to make it 0.
- There is only 1 input sequence for which the output of a Maxterm is 0. Because, it requires all values as zero.
- Then the product of all sum term (maxterm), from a function, and functions will have a value 0, if any of the sum term(maxterm) is 0.

| Binary Representation | Sequence | Maxterm        | Designation |
|-----------------------|----------|----------------|-------------|
| 000                   | 0        | $a + b + c$    | $M_0$       |
| 001                   | 1        | $a + b + c'$   | $M_1$       |
| 010                   | 2        | $a + b' + c$   | $M_2$       |
| 011                   | 3        | $a + b' + c'$  | $M_3$       |
| 100                   | 4        | $a' + b + c$   | $M_4$       |
| 101                   | 5        | $a' + b + c'$  | $M_5$       |
| 110                   | 6        | $a' + b' + c$  | $M_6$       |
| 111                   | 7        | $a' + b' + c'$ | $M_7$       |



## Canonical logic forms

- **Canonical POS form**: - In a product of sum form expression, if each OR term (sum term) consists all the literals(variables) appearing either in complements or uncomplemented form. E.g.  $(a' + b + c) \cdot (a + b' + c') \cdot (a + b + c)$ . Then the form is said to be Canonical POS form.

| Binary Representation | Sequence | Maxterm        | Designation |
|-----------------------|----------|----------------|-------------|
| 000                   | 0        | $a + b + c$    | $M_0$       |
| 001                   | 1        | $a + b + c'$   | $M_1$       |
| 010                   | 2        | $a + b' + c$   | $M_2$       |
| 011                   | 3        | $a + b' + c'$  | $M_3$       |
| 100                   | 4        | $a' + b + c$   | $M_4$       |
| 101                   | 5        | $a' + b + c'$  | $M_5$       |
| 110                   | 6        | $a' + b' + c$  | $M_6$       |
| 111                   | 7        | $a' + b' + c'$ | $M_7$       |

**Q** Given the function  $F = P' + QR$ , where  $F$  is a function in three Boolean variables  $P$ ,  $Q$  and  $R$  and  $P' = \neg P$ , consider the following statements. **(GATE-2015) (2 Marks)**

~~S1:  $F = \Sigma(4, 5, 6)$~~

S2:  $F = \Sigma(0, 1, 2, 3, 7)$  ✓✓

S3:  $F = \Pi(4, 5, 6)$  ✓✓

~~S4:  $F = \Pi(0, 1, 2, 3, 7)$~~

Which of the following is true?

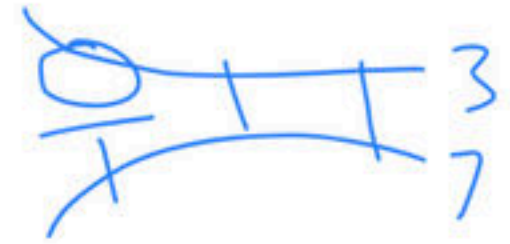
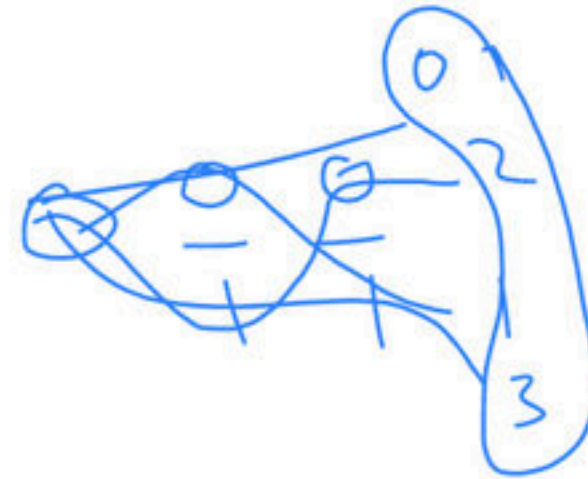
~~(A) S1-False, S2-True, S3-True, S4-False~~ — 63

(B) S1-True, S2-False, S3-False, S4-True — 20

(C) S1-False, S2-False, S3-True, S4-True — 8

(D) S1-True, S2-True, S3-False, S4-False — 9

$$f(P, Q, R) = \overline{P} + 0R$$



$$\frac{\Sigma(0, 1, 2, 3, 7)}{\Pi(4, 5, 6)} \leftarrow \begin{matrix} 1 \\ 0 \end{matrix}$$

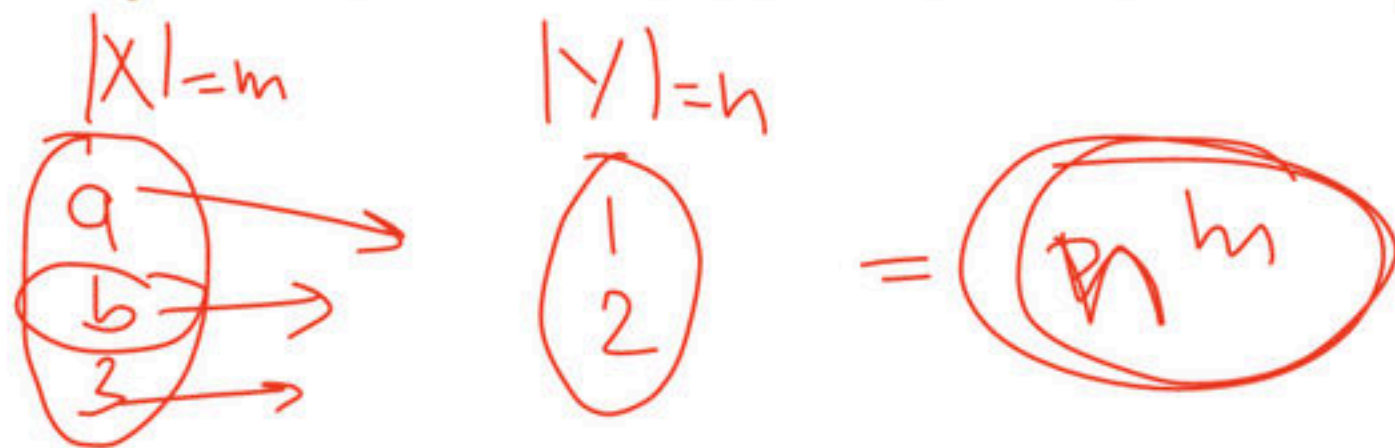


# Break

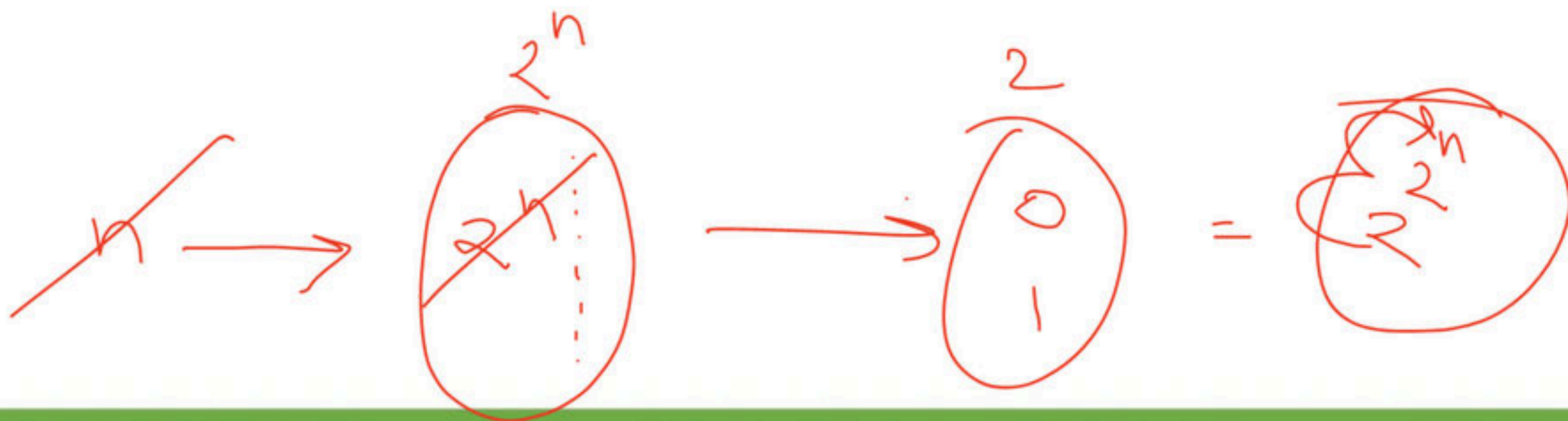
## No of functions possible

Q With n-Boolean variables how many different Boolean functions are possible?

| a | b |
|---|---|
| 0 | 0 |
| 0 | 1 |
| 1 | 0 |
| 1 | 1 |



$$2 \times 2 \times 2 = 8$$



$a, b \rightarrow$

| $a$ | $b$ | $j_0$ | $j_1$ | $j_2$ | $j_3$ | ... | $j_{15}$ |
|-----|-----|-------|-------|-------|-------|-----|----------|
| 0   | 0   | 0     | 0     | 0     | 0     |     | 1        |
| 0   | 1   | 0     | 0     | 0     | 0     |     | 1        |
| 1   | 0   | 0     | 0     | 1     | 1     |     | 1        |
| 1   | 1   | 0     | 1     | 0     | 1     |     | 1        |

$$2 \rightarrow 2^2 \rightarrow 2^{2^2}$$

## No of functions possible

**Q** With  $n$ -Boolean variables how many different Boolean functions are possible?

Solution: with  $n$  binary variable we can generate  $2^n$  combinations. And as we know that a Boolean function can also have only 2 possible values either 0 or 1, so total number of different functions possible will be  $2^{(2^n)}$ .

Similarly, we can generalize this idea as  $x^{(y^z)}$  are the total number of functions possible,  $x$  is the nature of the function,  $y$  is the nature of the variable and  $z$  is the number of variables.



**Q** How many different Boolean functions of degree 4 are there? (**NET-JUNE-2013**)

**(A)**  $2^4$

**(B)**  $2^8$

**(C)**  $2^{12}$

**(D)**  $2^{16}$

**Q** What is the maximum number of different Boolean functions involving  $n$  Boolean variables? **(GATE-2007) (1 Marks)**

**a)**  $n^2$

**b)**  $2^n$

**c)**  $2^{2^n}$

**d)**  $2^{n^2}$

# Break

## Complementation

- Let us consider a function  $f(a, b, c, d, \dots, 0, 1, +, \cdot)$ , then the complement of the function is defined as  $f'(a', b', c', \dots, z', 1, 0, \cdot, +)$ .
- i.e. When all the variables are replaced by their compliments,  $0 \rightarrow 1, 1 \rightarrow 0$ , or  $\rightarrow$  and, and  $\rightarrow$  or, then we will be called them compliment of a function.

$0, 1, 2, 3$

$f(a, b) = \sum_m(1, 2)$

$A \quad A'$

$\bar{f}(a, b) = \sum_m(0, 3)$

$\bar{f}(a, b) = \prod_m(1, 2)$



$$f(a,b) = \overline{a}b + a\overline{b}$$

$$(\overline{a}b) \cdot (a\overline{b})$$

$$(\overline{a} + b) \cdot (a + \overline{b})$$

$$\underline{(a + \overline{b})} \cdot \underline{(\overline{a} + b)}$$

$$\cancel{a \cdot a} + ab + \overline{b} \overline{a} + \cancel{\overline{b} \cdot b}$$

$$\rightarrow ab + \overline{a}\overline{b}$$

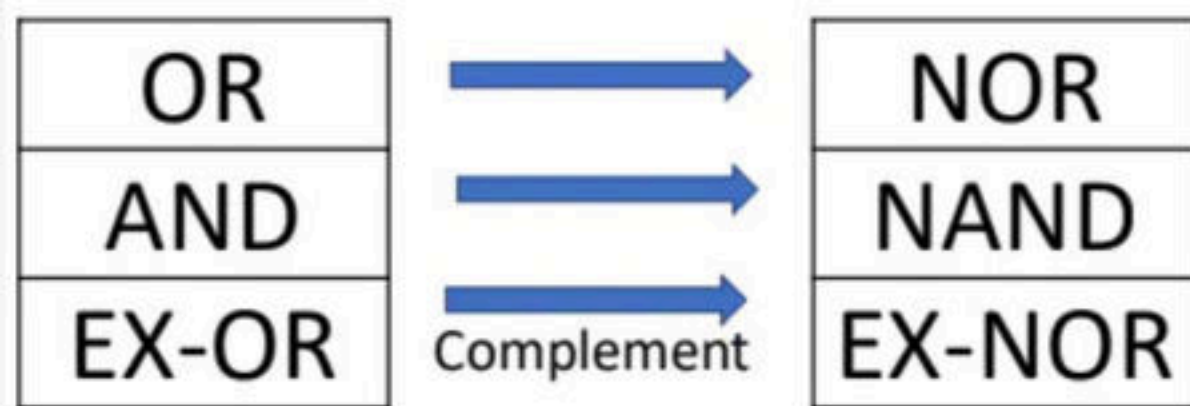
$$\overline{a} + ab\overline{c} + a\overline{b}\overline{c}d$$

$$\rightarrow a \cdot (\overline{a} + \overline{b} + c) \cdot (\overline{a} + b + c + d)$$

$$\overline{a}b + a\overline{b}$$

$$\underline{(a + \overline{b})} \cdot \underline{(\overline{a} + b)}$$

- We can directly put a bar of the entire Boolean expression to find complement of the function.
- We can also directly deal in minterms and maxterm to write complement function.



# Break

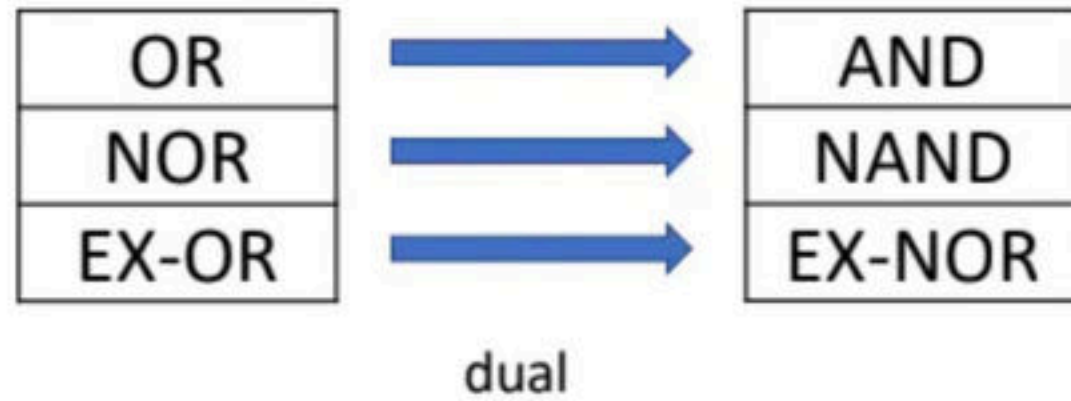
## Duality

- Let us consider a function  $f(a, b, c, d, \dots, z, 0, 1, +, \cdot)$ , then the dual of the function is defined as  $f^d(a, b, c, \dots, z, 1, 0, \cdot, +)$ .
- When the nature of variable remains same but  $0 \rightarrow 1$ ,  $1 \rightarrow 0$ , or  $\rightarrow$  and, and  $\rightarrow$  or, then they are called dual functions.

| a     | b     |       |       |
|-------|-------|-------|-------|
| L O H | L O H | L O H | L O H |
| L O H | H 1 L | H 1 L | L O H |
| H 1 L | L O H | H 1 L | L O H |
| H 1 L | H 1 L | H 1 L | H 1 L |



- When we take dual of a function, then the functionality of a function remains the same but a positive logic system is transformed to negative logic system.
- If a function works correctly in positive logic system, then it must also work correct in negative logic system as it does not depend on magnitude.



**Q** The dual of a Boolean expression is obtained by interchanging **(NET-DEC-2013)**

- (A)** Boolean sums and Boolean products
- (B)** Boolean sums and Boolean products or interchanging 0's and 1's
- (C)** Boolean sums and Boolean products and interchanging 0's & 1's
- (D)** Interchanging 0's and 1's

**Q** The dual of the switching function  $x + yz$  is: **(NET-DEC-2007) (NET-DEC-2008)**

**(A)**  $x + yz$

**(B)**  $x' + y'z'$

**(C)**  $x(y + z)$

**(D)**  $x'(y' + z')$

# Break



## Neutral Function

- A function  $f$  is said to be neutral if, it has equal number of minterms and maxterms.

## Self-Dual

- A function  $f$  is said to be self-dual if both the functions and its dual are same.
- $f = f^D$
- $F(a, b, c) = ab + bc + ca$

**Q** which of the following functions are self-dual?

$$f(a, b, c) = \sum_m (0, 3)$$

$$F(a, b, c) = \sum_m (0, 1, 6, 7)$$

$$F(a, b, c) = \sum_m (0, 1, 2, 4)$$

$$F(a, b, c) = \sum_m (3, 5, 6, 7)$$

**Q how to check weather a function is self-dual or not?**

1) check whether function is neutral or not i.e. (minterm = maxterm)

2) A self-dual function does not have any mutually exclusive terms. So in every pair of mutual exclusion term, we can pick only 1 minterm.

3) for  $n$  variable functions total  $2^n$  minterms are possible, so we will have  $2^{n-1}$  pair of mutually exclusive minterms, in every pair of mutually exclusive minterms we have two choice so, For  $n$  variable function total number of  $2^{(2^{n-1})}$  self-dual functions are possible.

$0 \leftrightarrow 7$

$1 \leftrightarrow 6$

$2 \leftrightarrow 5$

$3 \leftrightarrow 4$

e.g. of Mutual exclusive min terms



**Q** The dual of a Boolean function  $F(X_1, X_2, \dots, X_n, +, *, ')$ , written as  $F^D$ , is the same expression as that of  $F$  with  $+$  and  $*$  swapped.  $F$  is said to be self-dual if  $F = F^D$ . The number of self-dual functions with  $n$  Boolean variables is. **(GATE-2014) (1 Marks)**

- a)**  $2^n$                       **b)**  $2^{(n-1)}$                       **c)**  $2^{(2014^n)}$                       **d)**  $2^{(2^{(n-1)})}$

**Q** How many Boolean functions of degree  $n$  are self-dual? **(NET-JUNE-2014)**

- (A)**  $2^n$                       **(B)**  $(2)^{2n}$                       **(C)**  $2^{n^2}$                       **(D)**  $(2)^{\wedge (2^{n-1})}$

# Break

## Orthogonal

- A function  $f$  is said to be orthogonal if the compliment and dual of the function are same.
- $f' = f^d$
- $f(a, b, c) = a'b'c' + abc + a'bc' + ab'c$



- A function can never be both orthogonal and self-dual because it will imply the function is same to its compliment, which is never possible.

**Q** how to check whether a function is orthogonal or not?

1) check whether function is neutral or not i.e. (minterm = maxterm)

2) we can choose any pair of mutual exclusive terms, but the minterms and its complement, both must be present.

3) If there is a function  $f$  of  $n$ -variables, so we will have  $2^{n-1}$  pair of mutually exclusive minterms, out of which we have to select exactly half of the pairs ( $2^{n-2}$ ), then  $(2^{n-1})C(2^{n-2})$  different orthogonal functions are possible.

$0 \leftrightarrow 7$

$1 \leftrightarrow 6$

$2 \leftrightarrow 5$

$3 \leftrightarrow 4$

e.g. of Mutual exclusive min terms

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**Q.39** Consider a Boolean function  $f(w, x, y, z)$  such that

$$f(w, 0, 0, z) = 1$$

$$f(1, x, 1, z) = x + z$$

$$f(w, 1, y, z) = wz + y$$

The number of literals in the minimal sum-of-products expression of  $f$  is \_\_\_\_\_.